Notes: Tabellini (RES, 2020)

Gifts of the Immigrants, Woes of the Natives: Lessons from the Age of Mass Migration

Contents

1 Big picture

- **Question.** What were the political and economic effects of European immigration to U.S. cities during the Age of Mass Migration, and why did immigration generate political backlash?
- **Setting.** The paper studies 180 U.S. cities in 1910, 1920, and 1930, exploiting World War I and the Immigration Acts of 1921 and 1924.
- **Political result.** Immigration reduced public goods provision and taxes, reduced support for Democrats, increased conservative representation, and raised support for the National Origins Act.
- **Economic result.** Immigration increased native employment, occupational scores, industrial value added, and establishment size.
- **Culture result.** Backlash was strongest for culturally distant immigrants, especially non-Protestant and linguistically distant groups.
- **Takeaway.** Diversity can be economically productive and politically hard to manage.

2 Historical background, data, and measurement

2.1 Age of Mass Migration and policy shocks

Between 1850 and 1915, more than 30 million Europeans moved to the United States. World War I and the 1921/1924 Immigration Acts sharply changed both the number and the composition of immigrant flows. These national origin-specific shocks are central to the paper's identification strategy.

Interpretation: The setting combines large immigrant shares, changing cultural composition, and major policy shocks.

2.2 City sample and outcomes

The baseline sample is a balanced panel of 180 cities with at least 30,000 residents in 1910, 1920, and 1930. The paper measures demographics, local fiscal policy, presidential vote shares, congressional ideology, native employment, occupational scores, and manufacturing outcomes.

Interpretation: The same empirical design is used to study both politics and economics, which allows the paper to distinguish backlash from economic harm.

2.3 Cultural distance measures

The paper uses religion and linguistic distance. Religion separates Protestant from non-Protestant sending countries. Linguistic distance is a weighted average of each immigrant group's distance from English.

Interpretation: These variables test whether backlash depends on the type of immigrants rather than only on the number of immigrants.

3 Models and empirical specifications

3.1 Baseline equation

$$y_{cst} = \gamma_c + \delta_{st} + \beta Imm_{cst} + u_{cst}.$$

Here y_{cst} is an outcome, γ_c are city fixed effects, δ_{st} are state-by-year fixed effects, and Imm_{cst} is recent immigrants over predicted city population.

Interpretation: The coefficient β is identified from within-city changes in immigration relative to other cities in the same state-year.

3.2 Leave-out shift-share instrument

$$Z_{cst} = \frac{1}{\widehat{P}_{cst}} \sum_j \alpha_{jc} O_{jt}^{-M}.$$

α_{jc} is the 1900 ethnic settlement share; O_{jt}^{-M} is origin-country immigration net of the city's own MSA; $\widehat{P}_{cst}$ is predicted city population.

Interpretation: The instrument combines historical enclaves with aggregate origin shocks while excluding the city's own MSA inflow.

3.3 Religion and linguistic distance

$$y_{cst} = \gamma_c + \delta_{st} + \beta_1 Imm_{cst}^{NonProt} + \beta_2 Imm_{cst}^{Prot} + u_{cst}.$$

$$LD_{cst} = \sum_j share_{jcst} L_j.$$

Interpretation: These specifications ask whether cultural composition predicts political responses conditional on immigration levels.

4 Identification and empirical design

The main threat is that immigrants choose cities with changing economic or political conditions. The paper addresses this using a leave-out shift-share instrument, city fixed effects, state-by-year fixed effects, predicted population denominators, pre-trend tests, controls for predicted industrialization, interactions with 1900 city characteristics, alternative instruments based on WWI and quotas, MSA aggregation checks, and outlier analyses.

5 Section-by-section study guide

- Section 1 introduces the joint puzzle: immigration is economically beneficial but politically costly.
- Section 2 explains historical shocks and changing immigrant composition.
- Section 3 defines city, fiscal, political, economic, and cultural variables.
- Section 4 presents the baseline equation, instrument, and first-stage evidence.
- Sections 5–7 show political backlash, economic gains, and cultural mechanisms.
- Section 8 summarizes robustness checks.

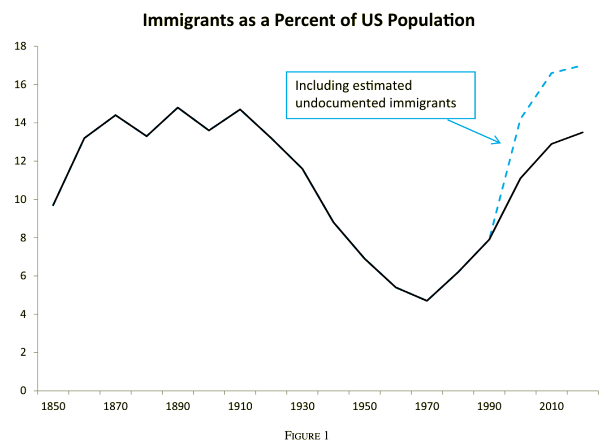
6 Tables and figures

6.1 Figure 1: Immigrants as a percent of U.S. population + Figure 2: Share of foreign born in the U.S., by region

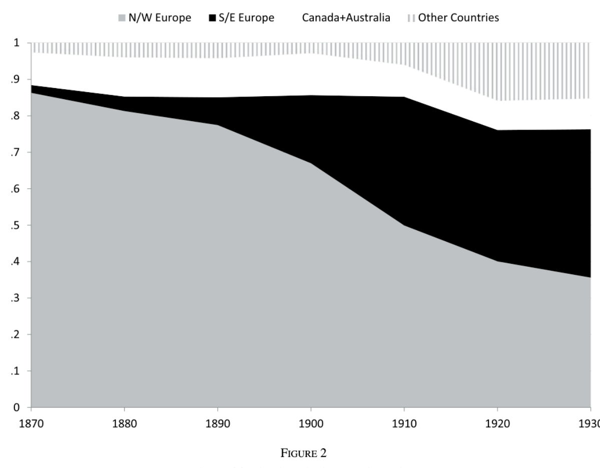
Read: Figure 1 shows that foreign-born shares were historically high. Figure 2 shows the shift from Northern/Western to Southern/Eastern European immigration.

Detailed interpretation: These figures motivate both the relevance of the historical episode and the importance of immigrant composition.

Economic message: The empirical design exploits large immigration and sharp compositional change.



(a) Immigrants as a percent of U.S. population



(b) Share of foreign born in the U.S., by region

6.2 Figure 3: Total number of immigrants

Read: The figure shows sharp disruptions from World War I and quota restrictions.

Detailed interpretation: These shocks generate the time-series component of the instrument.

Economic message: National shocks interact with local ethnic enclaves to generate city-level immigration variation.

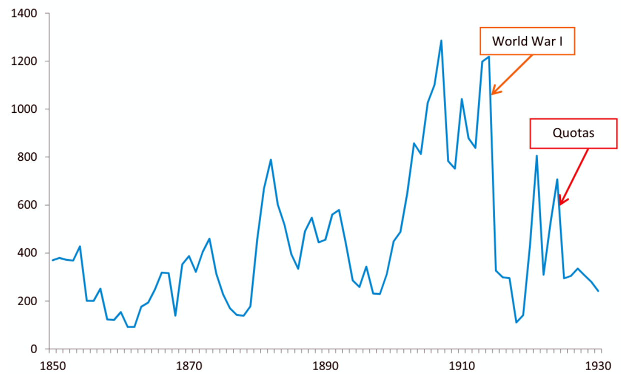


FIGURE 3
Total number of immigrants (in thousands)

economic impact of European immigrants, historian Maldwyn Jones wrote that "The
ization of America's vast economic potential has...been due in significant measure to the

Total number of immigrants in thousands

6.3 Table 1: Summary statistics + Table 2: First stage

Read: Table 1 summarizes the city panel. Table 2 shows a strong first stage, with coefficients close to one and high F-statistics.

Detailed interpretation: The first stage is strong across alternative scaling and robustness specifications.

Economic message: The instrument predicts actual immigrant inflows well.

TABLE 1
Summary statistics

VARIABLES	*Mean*	*Median*	*St. Dev.*	*Min*	*Max*	*Obs.*
Panel A. City demographics						
Fr. all immigrants	0.152	0.149	0.097	0.007	0.518	540
Fr. recent immigrants	0.042	0.026	0.044	0.001	0.343	540
Recent immigrants over 1900 population	0.074	0.048	0.078	0.002	0.678	540
City population (1,000s)	190.1	76.05	510.4	30.20	6,930	540
Panel B. Economic outcomes						
Employed	0.858	0.889	0.071	0.648	0.952	538
Log occupational scores	3.263	3.265	0.047	3.080	3.427	538
Value added per establishment	87.66	65.92	74.47	7.945	556.3	525
Establishment size	52.86	43.09	37.98	5.465	229.9	525
Panel C. Political outcomes						
Tax rate per 1,000$ of assessed valuation	29.42	25.78	16.48	6.450	114.3	539
Expenditures per capita	14.57	12.89	7.336	3.443	49.99	540
Democrats' vote share	0.482	0.465	0.189	0.103	0.967	378
DW nominate score	0.178	0.334	0.338	−0.578	0.991	470

Notes: The sample includes a balanced panel of the 180 U.S. cities with at least 30,000 residents in each Census year 1910, 1920, and 1930. *Employed* is the employment to population ratio for native men in the age range (15–65). *Fr. all immigrants* (respectively *Fr. recent immigrants*) is the total number of European immigrants (respectively the number of European immigrants arrived in the last 10 years) divided by city population.

(a) Summary statistics

TABLE 2
First stage

	Dep. variable: *fraction of immigrants*					
	(1)	(2)	(3)	(4)	(5)	(6)
Z	0.840***	0.968***	0.999***	0.948***	0.893***	0.900***
	(0.056)	(0.064)	(0.059)	(0.104)	(0.091)	(0.081)
1900 population		X				
Predicted population			X			
MSA analysis				X		
Year by 1900 log					City and imm pop	Value added manuf.
F-stat	225.1	226.7	288.3	82.65	96.48	124.8
Cities	180	180	180	127	180	176
Observations	540	540	540	381	540	528

Notes: The sample includes a balanced panel of the 180 U.S. cities with at least 30,000 residents in each Census year 1910, 1920, and 1930. In column 1, the actual number of immigrants is scaled by actual population, and the instrument is the leave-out version of the shift-share IV in equation (1) (Section 4.2). Columns 2 and 3 replicate column 1 by scaling the actual and predicted number of immigrants by, respectively, 1900 and predicted population. From column 3 onwards, this table presents results from specifications where both the predicted and the actual number of immigrants are scaled by predicted population. Column 4 replicates the analysis aggregating the unit of analysis at the MSA level. Columns 5 and 6 include the interaction between year dummies and, respectively, the (log of) 1900 city and immigrants population, and the (log of) 1904 value added by manufacture per establishment. F-stat refers to the K–P F-stat for weak instrument. All regressions partial out city and state by year fixed effects. Robust standard errors, clustered at the MSA level, in parenthesis. *** $p<0.01$ ** $p<0.05$ * $p<0.1$.

(b) First stage

6.4 Figure 4: First stage actual versus predicted immigration

Read: The residualized scatterplot shows a strong positive relationship between predicted and actual immigrant shares.

Detailed interpretation: Passaic, New Jersey is a visible outlier, but dropping it does not eliminate the first-stage relationship.

Economic message: The core empirical variation is visible graphically.

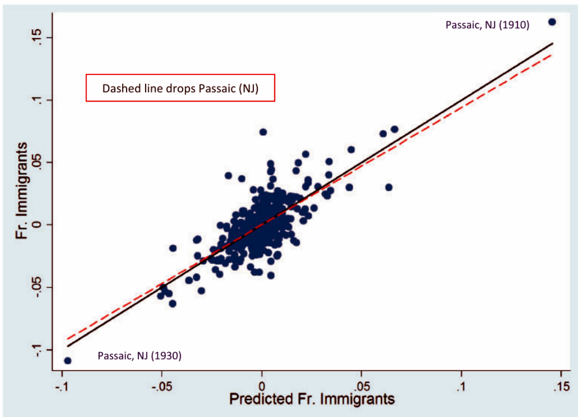


FIGURE 4
First stage: actual versus predicted immigration

TABLE 3
The political effects of immigration

Dependent variable	(1) Public spending per capita	(2) Property tax rate	(3) Democrats' vote share	(4) DW nominate score
Panel A: OLS				
Fr. immigrants	−5.958*	−28.49***	−0.528***	0.745
	(3.589)	(9.754)	(0.106)	(0.468)
Panel B: 2SLS				
Fr. immigrants	−8.699*	−29.44*	−0.404***	1.658**
	(4.453)	(16.95)	(0.141)	(0.808)
F-stat	288.3	292.7	83.14	23.11
Mean of dep var.	12.16	19.75	0.490	0.165
Observations	540	539	378	460

Notes: Columns 1 and 2 present results for a balanced panel of the 180 U.S. cities with at least 30,000 residents in each Census year 1910, 1920, and 1930. The dependent variable is public spending per capita in column 1 and property tax rate for $1,000 of assessed valuation in column 2. In column 3, the dependent variable is the Democrats vote share in Presidential elections, and the sample includes the balanced panel of the 126 metropolitan statistical areas (MSAs) containing at least one of the 180 cities in my sample. In column 4, the dependent variable is the first dimension of DW Nominate scores for members of the House of Representatives, for the panel of city-to-congressional district units for Congress 61, 66, and 71, for the 180 cities considered in my sample. Panels A and B present OLS and 2SLS results for the baseline specification (equation (1)). *Fr. Immigrants* is the fraction of immigrants arrived in the previous decade over predicted city population, and is instrumented using the baseline version of the instrument constructed in Section 4.2 (see (2) in the main text). F-stat refers to the K–P F-stat for weak instrument. Regressions in columns 1 and 2 include city and state by year fixed effects, while regressions in columns 3 and 4 include MSA (column 3) or congressional district to city (column 4) and state by year fixed effects. Robust standard errors, clustered at the MSA (for columns 1–3) or congressional district to city (column 4) level, in parenthesis. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$.

6.5 Table 3: The political effects of immigration

Read: Immigration reduces public spending, property tax rates, and Democratic vote share, and raises DW-NOMINATE scores.

Detailed interpretation: A five-percentage-point increase in immigration reduces spending by about 5%, tax rates by about 7.5%, and Democrats' vote share by about two percentage points.

Economic message: Immigration produces concrete political backlash and policy change.

6.6 Table 4: Congressmen ideology and the National Origins Act of 1924

Read: Immigration raises conservative representation and support for immigration restrictions.

Detailed interpretation: The effect is strongest for conservative Republicans and for votes supporting the National Origins Act.

Economic message: Immigration influenced national policy through representation.

TABLE 4
Congressmen ideology and the National Origins Act of 1924

Dep. Variable:	DW nominate scores		Pr. that winner has given political orientation				1[restrict immigration]	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: OLS								
Fr. Immigrants	0.745	0.603	−0.045	−0.804	−0.290	1.238	2.121*	2.024
	(0.514)	(0.521)	(0.317)	(0.711)	(0.991)	(1.135)	(1.189)	(1.362)
Panel B: 2SLS								
Fr. Immigrants	1.658**	1.575*	−0.601	−1.655	−0.198	2.592*	3.784**	3.365*
	(0.808)	(0.841)	(0.817)	(1.039)	(1.717)	(1.354)	(1.569)	(1.770)
F-stat	23.11	19.56	23.11	23.11	23.11	23.11	88.05	39.34
Mean dep var.	0.165	0.150	0.167	0.161	0.359	0.314	0.676	0.676
Observations	470	437	470	470	470	470	155	155
Balanced panel		X						
Political orientation			Liberal Democrat	Moderate Democrat	Moderate Republican	Conservative Republican		

Notes: Columns 1–6 report results for the panel of city-to-congressional district units for Congress 61, 66, and 71, for the 180 cities considered in my sample (see Supplementary Table A2). Because of redistricting between the 61st and the 66th Congress, it was not possible to construct a balanced panel including all city-congressional district cells in my sample. For this reason, column 2 restricts the attention to the balanced panel of cities (to congressional districts) that were not affected by redistricting. The unbalanced (respectively balanced) panel is composed of 157 (respectively 146) units of observations. Columns 7 and 8 present results from a cross-sectional regression for the 155 combinations of cities to congressional districts in Congress 68, for the 180 cities considered in my sample. Panels A and B report, respectively, OLS and 2SLS results. The dependent variable is the first dimension of the DW Nominate score in columns 1 and 2, an indicator for electing a politician with a given political orientation (see bottom of the table) in columns 3–6, and an indicator for voting in favour of the 1924 National Origins Act in the House of Representatives. *Fr. Immigrants* is the fraction of immigrants arrived in the previous decade over predicted city population, and is instrumented using the baseline version of the instrument constructed in Section 4.2 (see (2) in the main text). F-stat refers to the K–P F-stat for weak instrument. Columns 1–6 include city by congressional district and state by year fixed effects. Columns 7 and 8 control for state fixed effects. Column 8 also includes the 1900 log of black, immigrants, and total population, as well as

TABLE 5
The economic effects of immigration

	Natives' outcomes		Economic activity	
Dependent variable:	(1) Employment to population ratio	(2) Log occupational scores	(3) Log value added per establishment	(4) Log establishment size
Panel A: OLS				
Fr. Immigrants	0.287***	0.000	2.057***	2.195***
	(0.040)	(0.049)	(0.647)	(0.565)
Panel B: 2SLS				
Fr. Immigrants	0.299***	0.097***	2.889***	2.532***
	(0.064)	(0.036)	(0.954)	(0.815)
F-stat	251.3	251.3	270.5	270.5
Mean of dep var.	0.911	3.245	3.820	3.539
Observations	538	538	525	525

Notes: This table presents results for a balanced panel of the 180 U.S. cities with at least 30,000 residents in each Census year 1910, 1920, and 1930, restricting the attention to native men in the age range 15–65 who are not enrolled in schools (columns 1 and 2). Columns 3 and 4 further restrict the sample to city-year observations for which data were reported in the Census of Manufacture between 1909 and 1929. The dependent variable is natives' employment to population ratio in column 1, and natives' log occupational scores in column 2. Occupational scores are computed by IPUMS, and assign to an individual the median income of his job category in 1950. The dependent variable is the log of value added per establishment in column 3, and the log of establishment size in column 4. Panels A and B present OLS and 2SLS results for the baseline specification (equation (1)). *Fr. Immigrants* is the fraction of immigrants arrived in the previous decade over predicted city population, and is instrumented using the baseline version of the instrument constructed in Section 4.2

6.7 Table 5: The economic effects of immigration

Read: Immigration increases native employment, occupational scores, value added, and establishment size.

Detailed interpretation: These results contradict the idea that backlash came from direct economic losses to natives.

Economic message: The political backlash occurred despite positive local economic effects.

6.8 Table 6: Immigration and religion

Read: Non-Protestant immigration reduces redistribution and increases anti-immigration politics; Protestant immigration does not.

Detailed interpretation: This result targets the cultural-backlash mechanism.

Economic message: Political hostility is strongest toward culturally distant immigrants.

TABLE 6
Immigration and religion

Dep. Var.	(1) Total tax revenues PC	(2) Property tax revenues PC	(3) Property tax rate	(4) Public spending PC	(5) Dem-Rep. margin	(6) Smith's pct. votes	(7) DW nominate scores	(8) 1 [restrict immigration]
Panel A: OLS								
Fr. Non-Prot.	−0.519	−0.449	−1.235***	−0.320*	−0.039***	−0.042***	0.035	0.099***
	(0.329)	(0.278)	(0.477)	(0.180)	(0.007)	(0.007)	(0.025)	(0.037)
Fr. Prot.	0.406	0.277	−0.077	0.154	0.023	0.025*	−0.009	−0.057
	(0.339)	(0.326)	(0.722)	(0.313)	(0.016)	(0.015)	(0.017)	(0.045)
Panel B: 2SLS								
Fr. Non-Prot.	−0.515*	−0.483*	−1.219*	−0.366**	−0.017**	−0.050***	0.063**	0.145***
	(0.306)	(0.284)	(0.649)	(0.183)	(0.009)	(0.008)	(0.030)	(0.053)
Fr. Prot.	0.193	0.067	−0.109	−0.007	−0.010	0.036	0.006	−0.066
	(0.399)	(0.351)	(1.120)	(0.250)	(0.013)	(0.024)	(0.030)	(0.078)
KP F-stat	26.37	26.37	26.23	26.37	37.89	35.87	32.16	23.74
F-stat (Non-Prot)	115.9	115.9	118.9	115.9	50.64	38.60	85.91	69.49
F-stat (Prot)	27.53	27.53	27.39	27.53	38.95	36.58	32.27	21.68
Mean of dep var	12.76	12.10	19.75	12.16	0.180	0.398	0.165	0.676
Observations	540	540	539	540	378	126	460	155

Notes: This table presents results for a balanced panel of the 180 U.S. cities with at least 30,000 residents in each Census year 1910, 1920, and 1930. The analysis is conducted at the MSA rather than at the city level, fixing boundaries using 1940 definitions in Cols 5 and 6, and at the city to congressional district level in columns 7 and 8. Panels A and B report, respectively, OLS and 2SLS results. The dependent variable is displayed at the top of each column. *1[Restrict Immigration]* (column 8) is an indicator for voting in favour of the 1924 National Origins Act in the House of Representatives. In columns 1–5 and in column 7, *Fr. Non-Prot.* (respectively *Prot.*) refers to the fraction of immigrants arrived in the previous decade

TABLE 7
Linguistic distance and redistribution

Dep. Var.	(1) Total tax revenues PC	(2) Property tax revenues PC	(3) Property tax rate	(4) Public spending PC	(5) Education	(6) Police	(7) Charities and Hospitals	(8) Sanitation
Panel A: OLS								
Ling. distance	−0.361*	−0.346	−1.485*	−0.213	−0.050	−0.032	−0.010	−0.045
	(0.205)	(0.212)	(0.840)	(0.160)	(0.060)	(0.021)	(0.039)	(0.029)
Panel B: 2SLS								
Ling. distance	−0.875*	−0.809*	−2.308	−0.519*	−0.199*	−0.013	−0.119	−0.053
	(0.468)	(0.458)	(1.598)	(0.301)	(0.117)	(0.042)	(0.084)	(0.052)
KP F-stat	21.02	21.02	21.47	21.02	21.14	21.02	16.31	21.02
F-stat (Imm.)	123.1	123.1	124.7	123.1	106.9	123.1	101.6	123.1
F-stat (Ling.)	50.38	50.38	53.48	50.38	48.05	50.38	34.06	50.38
Mean of dep var	12.76	12.10	19.75	12.16	4.250	1.338	0.635	1.129
Observations	540	540	539	540	534	540	516	540

Notes: This table presents results for a balanced panel of the 180 U.S. cities with at least 30,000 residents in each Census year 1910, 1920, and 1930. Panels A and B report, respectively, OLS and 2SLS results. The dependent variable is displayed at the top of each column. In columns 5–8, the dependent variable is spending per capita on the category listed at the top of the column. The main regressor of interest is the (standardized) weighted average linguistic distance constructed in Section 7.2, instrumented using predicted shares of immigrants from each sending region obtained from (2) in Section 4.2. F-stat is the Kleibergen–Paap F-stat for joint significance of instruments. F-stat (Imm.) and F-stat (Ling.) refer to the partial F-stats for joint significance of the instruments in the two separate first-stages. All regressions include the main effect of immigration (instrumented with the baseline shift-share instrument from (2)), and control for city and state by year fixed effects. Robust standard errors, clustered at the MSA level, in parenthesis. *** $p<0.01$ ** $p<0.05$ * $p<0.1$.

6.9 Table 7: Linguistic distance and redistribution

Read: Higher linguistic distance reduces tax revenues and public spending, especially education.

Detailed interpretation: Linguistic distance reinforces the religion evidence using a continuous measure of cultural distance.

Economic message: Culturally distant immigration reduces support for public goods and redistribution.

7 Result map

The results move in four steps. First, the instrument strongly predicts immigration. Second, immigration causes political backlash. Third, immigration produces positive economic outcomes, undermining the economic-threat explanation. Fourth, cultural distance explains heterogeneity in backlash.

8 Short summary

- The paper studies 180 U.S. cities during 1910–1930.
- Immigration reduces redistribution, support for Democrats, and liberal representation.
- Immigration increases native employment and industrial activity.
- Backlash is strongest for non-Protestant and linguistically distant immigrants.
- The evidence points to cultural/social-cohesion mechanisms rather than direct economic harm.

9 Conclusion

9.1 Core conclusion

Immigration during the Age of Mass Migration produced economic gains but political backlash. The backlash took the form of lower local redistribution, less Democratic support, more conservative representatives, and stronger support for national immigration restrictions.

9.2 Economic intuition

Immigrants supplied labour during a high-growth industrial period and helped firms expand. Native workers benefited through higher employment and occupational upgrading.

9.3 Political intuition

The backlash was concentrated among culturally distant immigrants, suggesting that identity, religion, language, and social cohesion mattered more than labour-market losses.

9.4 Policy implications

Economic gains from immigration do not automatically generate political support. Integration and social-cohesion policies may be essential complements to immigration policy.

9.5 Limitations

The paper is historical and cannot directly observe individual beliefs, but it provides unusually rich evidence linking economics, politics, and culture.

9.6 Takeaway

Immigration can be both a gift to the economy and a source of political conflict if diversity is socially hard to manage.